



S26-B10-Adv Data A-G11-E

Hotel Bookings Data Analysis Project

Project Documentation

Team 4

Name	Task
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September 20, 2026

Dataset Overview

The dataset used in this project is the **Hotel Booking Demand** dataset, containing booking records for a Resort Hotel and a City Hotel. Below is a description of each column as it appears in the raw data.

Column	Meaning
hotel	Type of hotel: Resort Hotel or City Hotel.
is_canceled	Whether the booking was canceled (1) or not (0).
lead_time	Number of days between the booking date and the arrival date.
arrival_date_year	Year of arrival date.
arrival_date_month	Month of arrival date (name).
arrival_date_week_number	Week number of the arrival date (1–53).
arrival_date_day_of_month	Day of the month of arrival.
stays_in_weekend_nights	Number of weekend nights (Sat/Sun) the guest stayed.
stays_in_week_nights	Number of week nights (Mon–Fri) the guest stayed.
adults	Number of adults in the booking.
children	Number of children in the booking.
babies	Number of babies in the booking.
meal	Type of meal package booked (BB, HB, FB, SC/Undefined).
country	Country of origin of the guest (ISO 3-letter code).
market_segment	Market segment designation (e.g. Direct, Corporate, Online TA).
distribution_channel	Booking distribution channel (e.g. Direct, TA/TO, Corporate).
is_repeated_guest	Whether the booking is from a repeat guest (1) or a new guest (0).
previous_cancellations	Number of previous bookings canceled by the customer before the current booking.
previous_bookings_not_canceled	Number of previous bookings not canceled by the customer before the current booking.
reserved_room_type	Code of the room type originally reserved.
assigned_room_type	Code of the room type actually assigned at check-in.
booking_changes	Number of changes/amendments made to the booking.
deposit_type	Deposit type: No Deposit, Non Refund, or Refundable.
agent	ID of the travel agency that made the booking.
company	ID of the company/entity responsible for paying for the booking (anonymized).
days_in_waiting_list	Number of days the booking was on the waiting list before confirmation.
customer_type	Type of booking arrangement: Transient, Transient-Party, Contract, or Group.
adr	Average Daily Rate – the average revenue earned per occupied room per night.
required_car_parking_spaces	Number of parking spaces required by the guest.
total_of_special_requests	Number of special requests made by the guest.
reservation_status	Last reservation status: Canceled, Check-Out, or No-Show.
reservation_status_date	Date at which the last reservation status was set.

1 Yousef Adel Abd ElShafy Ahmed: Python – Data Cleaning, Transformation & Visualization

This section summarizes the work I carried out on the Python side of the project: preparing the raw dataset for analysis and building the visualizations used across the dashboard pages.

1.1 Data Cleaning

- Removed exact duplicate rows from the dataset, since duplicated bookings would have inflated counts in every downstream chart.
- Standardized inconsistent missing-value markers (empty strings, "Nan", "None", "NA") into a single, proper null representation so pandas would recognize them consistently.
- Reviewed the missing-value profile of the dataset and handled each affected column individually, based on what a missing value actually meant for that specific field, rather than applying one blanket rule everywhere:
 - **Children:** a small number of missing values (negligible relative to dataset size) were removed, since they had no material effect on the analysis.
 - **Country:** rows with a missing country were removed, since country has no logical default value and guessing one (e.g. filling with the most common country) would have distorted every nationality-based chart.
 - **Agent:** missing values were filled with 0, since a missing agent ID reflects a genuine business meaning – no travel agency was involved in that booking (i.e. it was booked directly) – rather than an unknown value.
 - **Company:** missing values were filled with 0 for the same reason – a missing company ID means no company was paying for or responsible for the booking, not that the information was unavailable.
 - **Zero-night, non-canceled bookings:** a small number of rows had both `stays_in_weekend_nights` and `stays_in_week_nights` equal to 0 while `is_canceled` was False – meaning the booking was recorded as completed despite representing a stay of zero total nights, which is not logically possible for a real hotel stay (unlike a canceled booking with 0 nights, which is expected). These rows were identified and removed from the dataset, after confirming they represented a negligible percentage of the total data.

1.2 Data Transformation & Feature Engineering

- Converted key flag columns (`is_canceled`, `is_repeated_guest`) into proper boolean type for clarity and correctness in later grouping and filtering operations.
- Built a true, unified **arrival date** field by combining the year, month, and day-of-month columns, making time-based analysis and sorting far more reliable than working with the three separate columns.
- Added an **arrival_date_month_number** column (numeric month value) alongside the text month name, so that month-based charts could be sorted correctly in calendar order rather than alphabetical order.
- Created **has_agent** and **has_company** flag columns, derived from the agent/company ID fields, to give a simple, chart-friendly yes/no view of whether a booking involved a travel

agency or a company, separate from the detailed ID itself.

- Decoded abbreviated categorical fields into full, readable labels for use in chart legends and axis titles, while keeping the original short codes intact for filtering/grouping purposes:
 - `meal` → **meal_label** (e.g. BB → Bed & Breakfast, HB → Half Board, FB → Full Board, SC/Undefined → No Meal Package)
 - `market_segment` → **market_segment_label** (e.g. Online TA → Online Travel Agent)
 - `distribution_channel` → **distribution_channel_label** (e.g. TA/TO → Travel Agent / Tour Operator)
- Renamed the ambiguous `adr` column to **average_daily_rate(ADR)** for clarity throughout the analysis and all charts.
- Built a full country-code reference table (ISO 3166 alpha-2/alpha-3/country name) and merged it against the dataset's country codes to obtain full, readable **country names** for every booking, including manual handling of a legacy/deprecated code (TMP, the retired code for Timor-Leste) that would not have matched the current ISO standard.
- Created a **room mismatch** flag comparing the reserved room type against the assigned room type, to identify bookings where the guest did not end up in the room type they originally booked.

1.3 Key Performance Indicators

Metric	Value
Number of Bookings	86,323
Number of Hotels	2
Average Daily Rate (ADR)	107.34

1.4 Chart Count Summary

In total, I built **27 charts** across the six dashboard pages, using Matplotlib, Seaborn, and Plotly. The breakdown by page is:

Page	Charts
Home	2
Cancellations	5
Demand & Seasonality	4
Guest Behavior	5
Revenue & Pricing	5
Source & Operations	6
Total	27

The chart types used include line charts, bar charts, pie charts, a box plot, a scatter plot, and a choropleth map.

Using Matplotlib, Seaborn, and Plotly, I designed and built the chart set powering each page of

the dashboard:

1.5 Visualizations Built

Home page

- Monthly Booking Trend (line chart) – overall booking volume over the full period of the dataset.
- Cancellation Rate by Hotel Type (bar chart) – comparing Resort vs City hotel cancellation behavior.

Cancellations page

- Monthly Cancellation Trend (line chart).
- Cancellations by Month (line chart) – seasonal cancellation pattern across calendar months.
- Cancellation Rate by Customer Type (bar chart).
- Cancellation Rate by Deposit Type (bar chart).
- Cancellation Rate by Market Segment (bar chart).

Demand & Seasonality page

- Bookings by Month (line chart).
- Bookings by Arrival Day of Month (bar chart).
- ADR by Month (line chart).
- Bookings by Country (choropleth map).

Guest Behavior page

- Repeat vs New Guests (pie chart).
- Bookings by Guest Composition – adults/children/babies (bar chart).
- Agent-Booked vs Direct Guests (pie chart).
- Special Requests vs Cancellation Rate (bar chart).
- Company-Paid vs Individual Guests (pie chart).

Revenue & Pricing page

- ADR Trend by Month (line chart).
- ADR vs Lead Time (scatter plot).
- Average ADR by Customer Type (bar chart).
- Average ADR by Room Type (bar chart).
- ADR Distribution by Hotel Type (box plot).

Source & Operations page

- Bookings by Market Segment (pie chart).
- Bookings by Distribution Channel (bar chart).
- Reserved vs Assigned Room Type mismatch (pie chart).
- Meal Plan Popularity (pie chart).
- Top 15 Countries by ADR (bar chart).
- Guest Nationality – Top 15 (bar chart).

1.6 Key Insights Identified

- Cancellation likelihood varies noticeably by deposit type, market segment, and customer type – these are strong predictive signals for cancellation risk.
- Cancellation rate tends to decrease as the number of special requests increases, suggesting more engaged guests are less likely to cancel.
- ADR shows clear seasonal variation across months, and its variance increases substantially at longer lead times, reflecting fewer bookings (and therefore less pricing stability) far in advance.
- A measurable share of bookings show a mismatch between the room type reserved and the room type ultimately assigned, which is a useful operational quality indicator.
- Booking volume and guest nationality are heavily concentrated in a small number of source countries, with Portugal being the dominant market given the dataset's origin.

2 Asmaa Mohamed Mahmoud: Power BI – Dashboard Design & KPIs

This section summarizes the work carried out on the Power BI side of the project: cleaning the dataset in Power Query, defining the KPIs, and designing the suggested dashboard visualizations.

2.1 Project Objective

- Analyze hotel booking data using Microsoft Power BI.
- Clean the dataset and improve data quality before analysis.
- Identify booking patterns and cancellation behavior.
- Create clear KPIs and visualizations for management-style reporting.
- Present the results in a simple and interactive dashboard.

2.2 Dataset Overview

The project uses the `hotel_bookings.csv` dataset. The original dataset contains booking information covering hotel type, booking status, arrival dates, length of stay, guests, market segments, customer types, room types, daily rates, and reservation status.

Metric	Value
Original Rows	119,390
Columns	32
Exact Duplicate Rows	31,994
Columns Containing Nulls	4

2.3 Data Cleaning (Power Query)

The data was inspected and cleaned using Power Query in Power BI.

- Exact duplicate rows were removed to prevent the same booking from being counted more than once.
- Missing values in the **children** column were replaced with 0, since a blank value can reasonably represent no recorded children.
- Column names and data types were inspected before analysis.
- Missing values in **agent** and **company** were not automatically replaced, since a blank can represent that no agent or company was recorded for that booking.
- Unnecessary transformations were avoided in order to preserve the original information in the dataset.

2.4 Analysis Approach

1. Load the CSV dataset into Power BI.
2. Inspect the structure, missing values, duplicates, and data types.
3. Clean the data in Power Query.
4. Build KPIs and measures for the dashboard.
5. Use charts to compare hotels, cancellations, customer segments, and booking behavior.
6. Present the results in an interactive report.

2.5 Recommended Dashboard KPIs

KPI	Meaning
Total Bookings	Count of booking records.
Cancellation Rate	Canceled bookings divided by total bookings.
Average ADR	Average daily room rate.
Average Lead Time	Average number of days between booking and arrival.

These KPIs give a quick summary of booking volume, cancellation behavior, pricing, and booking lead time.

2.6 Suggested Visualizations

- Column chart: number of bookings by hotel type.
- Donut or column chart: canceled vs. non-canceled bookings.
- Bar chart: bookings by market segment.
- Bar chart: bookings by customer type.
- Column or line chart: booking trends by arrival year.
- Slicers: hotel, arrival year, market segment, and customer type.

2.7 Business Questions Addressed

- How many bookings are in the cleaned dataset?
- What proportion of bookings were canceled?
- Which hotel type receives more bookings?
- Which market segments generate the most bookings?
- How long in advance do customers usually book?
- How does the average daily rate differ across hotel types?
- Which customer groups appear most frequently?

2.8 Conclusion

Power BI provides an interactive way to explore the hotel booking dataset. Data cleaning is important because duplicates and missing values can affect KPIs and visualizations. The dashboard focuses on booking volume, cancellation behavior, pricing, and customer characteristics, allowing users to explore patterns and compare different booking segments.